

Multimodal Real Estate Price Prediction

Name – Suyash Chandrakant Patil

Branch – Production and Industrial Engineering

Enrollment – 24119040

Mobile – 9373187580

Linkedin - [Linkedin](#)

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GitHub link - [CDC x Yhills](#)

Overview

This project improves real estate price estimation by combining traditional property data with satellite images of the surrounding area. Instead of relying only on numbers, the model also captures neighbourhood quality and visual appeal, helping it understand factors like curb appeal that strongly influence property value.

1. Data Acquisition

A custom Python script was developed to retrieve over **21,000 satellite images(training + testing)**, each corresponding to precise geographic coordinates associated with individual properties.

2. Preprocessing

The tabular dataset was aligned with the available satellite images to ensure consistency across modalities. Logical validation checks, including standards derived from the **International Residential Code (IRC)**, were applied to detect and correct data anomalies.

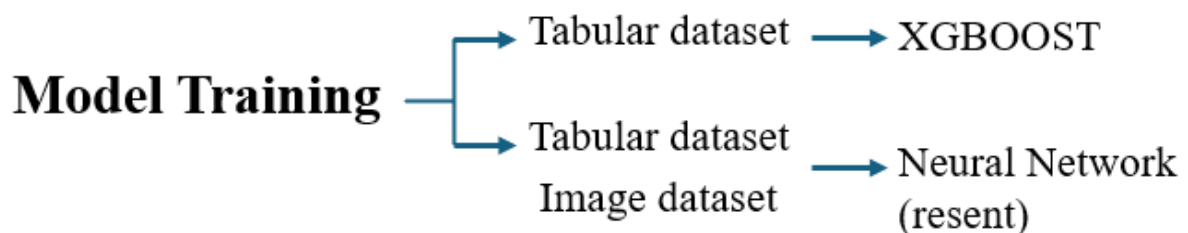
3. Scaling

To improve model stability and reduce skewness, house prices were first log-transformed and subsequently normalized to a **0–1 range**, ensuring better convergence during training.

4. Modelling

The final architecture consisted of two parallel neural networks. A **Convolutional Neural Network (CNN)** was used to extract spatial and visual features from satellite imagery, while a **Multi-Layer Perceptron (MLP)** processed tabular property attributes. Outputs from both networks were fused and passed to a final regression head for price prediction.

- **Tabular Stream:** Processes historical housing data—such as bedrooms, bathrooms, and square footage—using **XGBoost**.
- **Visual Stream:** Utilizes a **Convolutional Neural Network (CNN)** built with **TensorFlow** to extract high-dimensional visual embeddings from satellite images.
- **Fusion:** These two streams are connected to create a single predictive system that outputs the final property price

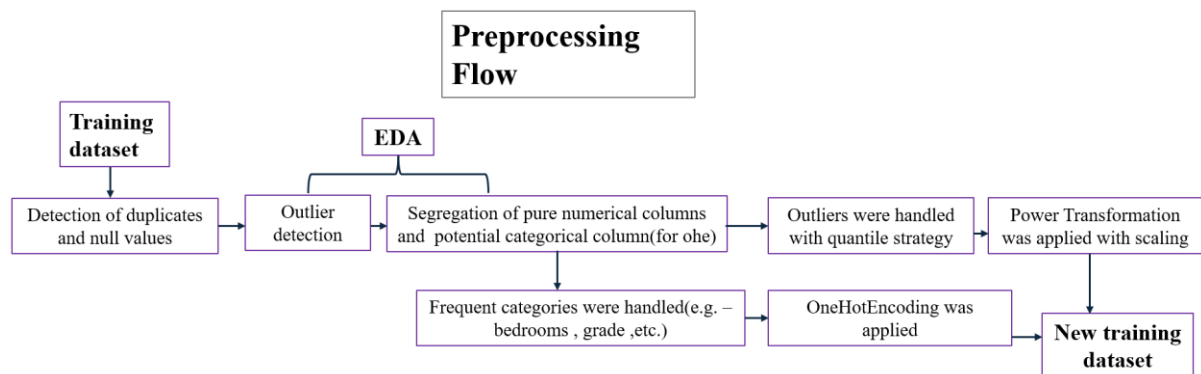


Prediction done using Neural Network Model

A pre-trained ResNet-50 architecture was employed to extract high-level spatial features from satellite imagery, enabling the model to capture complex visual patterns related to neighbourhood characteristics and environmental context.

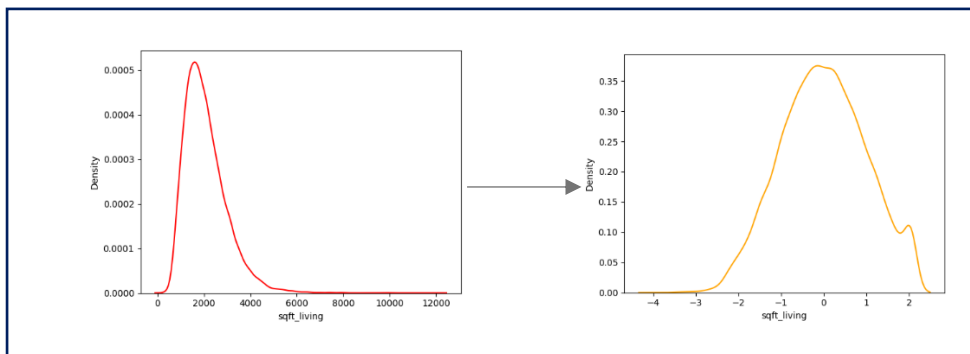
EDA

Exploratory Data Analysis (EDA) was conducted to understand data distributions, detect anomalies, and identify relationships between key features and property prices.

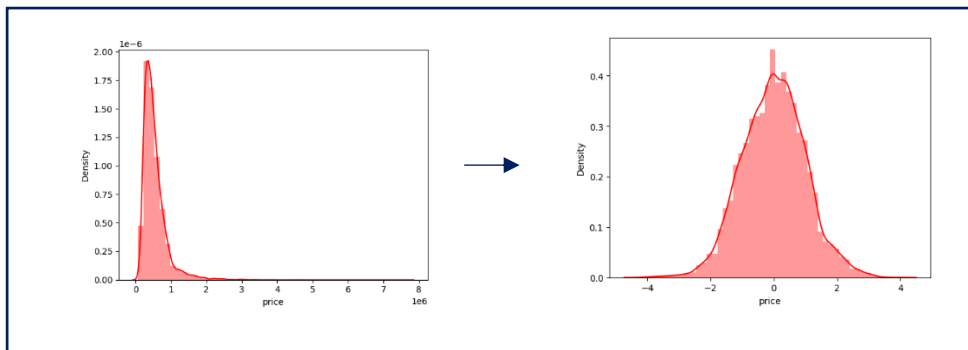


Pure numerical cols – sqft_living , sqft_lot ,etc.

Potential categorical – bathrooms ,grade ,waterfront ,etc.



Price distribution



The visualizations highlight strong right skewness key numerical features such as sqft_living and price. After applying power transformation, the distributions

become more symmetric and closer to normal, reducing the impact. This improves model stability and ability to learn data features

Analysis of the satellite images revealed substantial variation in visual density across properties. Locations with higher **greenery ratios** - such as the presence of trees and nearby parks - showed a statistically significant correlation with increased price per square foot. Notably, this relationship persisted even when key structural attributes, including `sqft_living`, were held constant.

Financial and Visual Insights

By integrating visual information from satellite imagery with traditional structural data, the model uncovered several key drivers influencing property valuation.

- **Visual Drivers:**

Satellite imagery analysis revealed that proximity to waterfronts and overall neighbourhood density had a significant impact on property prices. The convolutional neural network effectively differentiated between dense, concrete-dominated urban regions and foliage-rich suburban areas, with the latter consistently associated with premium property values. Green cover thus emerged as a strong visual proxy for neighbourhood desirability.

- **Structural Drivers:**

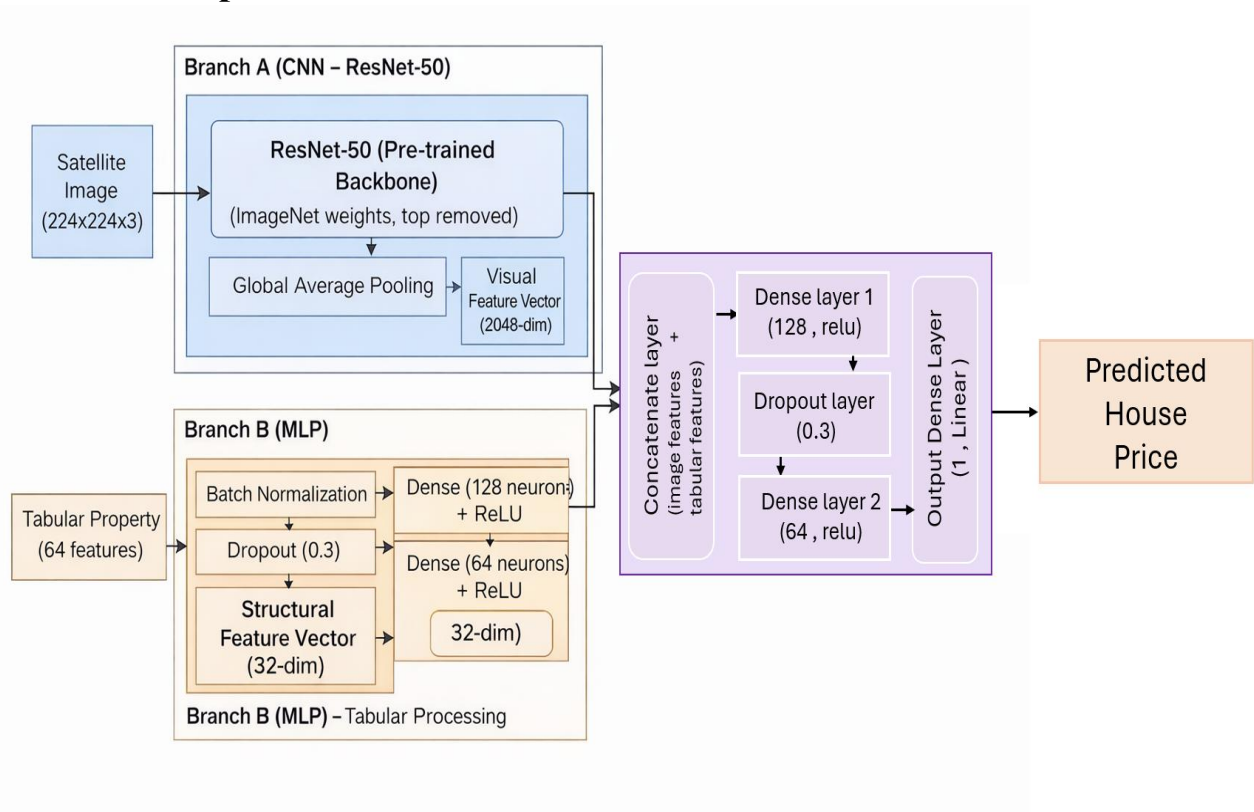
Analysis of tabular features indicated that construction grade and `sqft_living` remained the most influential structural predictors of property value, even after feature scaling and normalization.

- **Interaction Effects:**

The multimodal framework demonstrated strength in identifying interaction effects between visual and structural features. Specifically, it detected scenarios where high-quality homes located in low-greenery environments were valued lower than structurally similar properties in greener areas - an insight that is often overlooked by models relying solely on tabular data.

Architecture Diagram

A pretrained model ResNet-50 model was used for building neural network and prediction



ResNet-50 was used to extract visual features from satellite images, while an MLP was employed to capture patterns in tabular property data. The features from both branches were then combined to form a multimodal regression model for house price prediction.

This fusion enabled the model to jointly learn neighbourhood-level visual cues and structural property characteristics, resulting in improved predictive performance compared to unimodal approaches.

Result

- **XGBOOST** – 0.88 of cross_val_score after hype tuning
- **Neural Network** – A separate train and test splits were made from training data, and r2_scores were predicted on model trained on full training dataset

Test – 0.83, Mae - 0.2967(scaled data)

Train – 0.85, Mae – 0.2796(scaled data)

This ensures there is no overfitting in neural networks

Conclusion

Even though the dataset was relatively small for a deep model like ResNet-50, the proposed multimodal approach was able to achieve reliable prediction performance. By leveraging transfer learning, the model successfully extracted useful visual information from satellite images and combined it with tabular property data. This shows that incorporating visual context can meaningfully improve real estate price prediction, even when data availability is limited.

THANK YOU !!